

Extending Ab Initio Multiple Spawning with Ladder Operator Excitations

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Abstracts

Ab initio multiple spawning (AIMS) describes nonadiabatic dynamics with a basis of frozen Gaussians that is spawned near regions of nonadiabatic coupling. Because each basis function is a single frozen Gaussian, converging the population transfer through a conical intersection (CI) requires a large basis and many initial conditions, and even then transfer away from the CI is hard to capture. We extend AIMS by dressing each moving Gaussian with ladder operator excitations, so that the Hermite structure generated by the coupling is represented inside a frame rather than by additional spawns — one excited frame does the work of a dense cluster of spawned Gaussians. Spawning handles the transfer near the crossing (CI), where population builds up and separates onto the acceptor surface, while the ladder operators handle the transfer elsewhere, where the coupling only dresses the frame with co-moving excitations. On the four-mode pyrazine model, benchmarked against exact MCTDH, the method converges to the exact diabatic populations with a minimal basis of $N=183$ (maximum error 0.178) — about $2\times$ more accurate at 40% of the basis of adiabatic AIMS ($N=465$, error 0.329).

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Motivation

Nonadiabatic dynamics at conical intersections (CI) is critically important, and there is a need for methods that describe it both accurately and efficiently.

Single frozen Gaussian is exact for the dynamics on a harmonic potential, but it breaks down in two cases:

- Nonadiabatic coupling(adiabatic picture) or Electronic coupling(diabatic picture)
- Anharmonic potential

Ab initio multiple spawning (AIMS) tackles this by spawning basis functions.

However,

- Large basis is required to capture the population near the CI, many initial conditions (Wigner sampling) still fall short.
- AIMS spawns only where the NAC is large, missing transfer on anharmonic regions of the PES.

Introducing ladder operators

Regard a frozen Gaussian as the ground state $|0(q, p)\rangle$. Applying the raising operator generates excited states $|1(q, p)\rangle, |2(q, p)\rangle, \dots$ at the same center.

$$|n(q, p)\rangle = c \left(\hat{a}^\dagger \right)^n |0(q, p)\rangle \quad (1)$$

Why we introduce ladder operators?

- $|n(q, p)\rangle$ is finite difference of $\{|0(q_i, p_i)\rangle\}$.

$$\hat{a}^\dagger |0\rangle = \partial_\alpha \hat{D}(\alpha) |0\rangle \big|_{\alpha=0}, \quad |n\rangle = \frac{1}{\sqrt{n!}} \partial_\alpha^n (\hat{D}(\alpha) |0\rangle) \big|_{\alpha=0} \quad (2)$$

level $n = n$ -th derivative (finite difference) of a displaced Gaussian.

$\Rightarrow$ One $|n(q, p)\rangle$ captures inside a single frame what a dense cluster of spawns would.

- Diabatic picture : Linear electronic coupling is exactly captured by ladder operators.

$$\hat{V} = c\hat{q} \propto \hat{a}^\dagger + \hat{a} \quad (3)$$

- Adiabatic picture : Nonadiabaticity operator contains ladder operators.

$$\hat{\Theta}_{ab} = \frac{1}{2M} \left(2 \langle \phi_a | \nabla_n | \phi_b \rangle \nabla_n + \langle \phi_a | \nabla_n^2 | \phi_b \rangle \right), \quad \nabla_n \sim \hat{a}^\dagger - \hat{a} \quad (4)$$

Also, near the CI, PES is anharmonic.

Introducing ladder operators

Ladder operators capture the Hermite structure of the Hamiltonian without spawning.

⇒ The same accuracy is reached with fewer spawns and a smaller basis.

Therefore,

- Instead of one Gaussian per frame, we load a set of ground + excited state (excited Gaussians up to level K per mode) as the basis.
- They share the same center and width, move together along the classical trajectory, and their coefficients are propagated with the full-overlap variational metric.

Why adiabatic picture fails

However, ladder operators break the locality of the Gaussians.

⇒ If ladder operators are used, the adiabatic picture is no longer usable.

AIMS uses the saddle-point (centroid) approximation

$$\langle \chi_j | \hat{V} | \chi_k \rangle \simeq V(\bar{q}_{jk}) \langle \chi_j | \chi_k \rangle \quad (5)$$

Once locality is lost, this approximation breaks down.

⇒ So we must evaluate $\langle \chi_j | \hat{V} | \chi_k \rangle$ over regions where the NAC acts.

⇒ In the adiabatic picture the NAC diverges there.

⇒ the matrix element cannot be computed.

Use the diabatic picture.

- On the fly: local diabatization with a local harmonic approximation (LHA).
- In the diabatic picture the coupling is smooth (linear in q), so the matrix element stays finite (indeed analytic) even for delocalized ladder states.

Spawning in the diabatic picture

Then, how do we spawn in the diabatic picture?

- In the diabatic picture, electronic transitions occur *everywhere*; spawning at every point would make the basis far too large.
- Transitions are strongest near the crossing of the diabatic surfaces ($\approx$ the conical intersection).

So we split the roles:

- Transitions near the crossing $\rightarrow$ handled by **spawning**.
- Transitions elsewhere $\rightarrow$ handled by the **ladder operators**.

A few details:

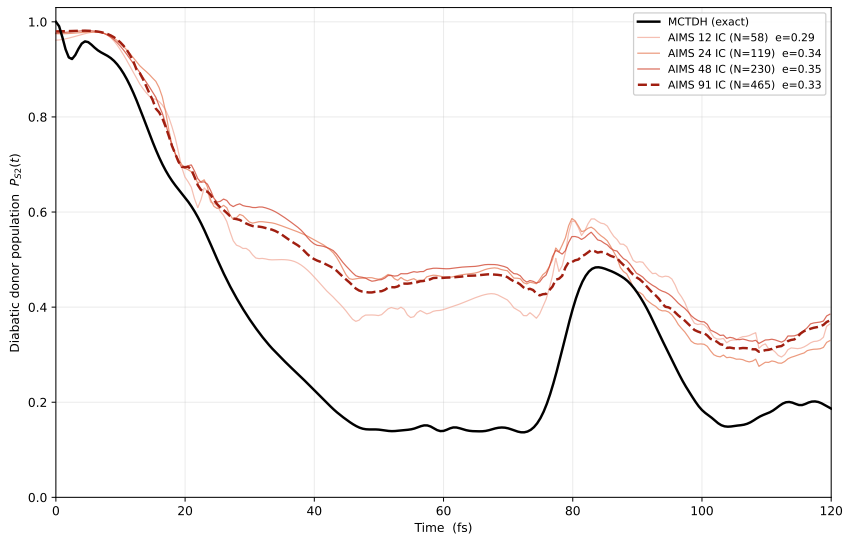
- Giving every frame $\{|0(q_i, p_i)\rangle\}$ the same excitation level K is wasteful — the acceptor frames use a slightly lower excitation.
- With many modes the number of excitation combinations explodes ($K_1 \times K_2 \times \dots$), so we keep only a selected subset.

Results: Computational details

- We use the four-mode pyrazine model from the Heidelberg MCTDH examples (pyrmod4.op).
- The exact populations are computed with Heidelberg MCTDH program.
- The AIMS populations are computed with pySpawn17 in the adiabatic representation and then converted to diabatic populations.
- Ladder operator populations are computed with my own python code in the diabatic representation.

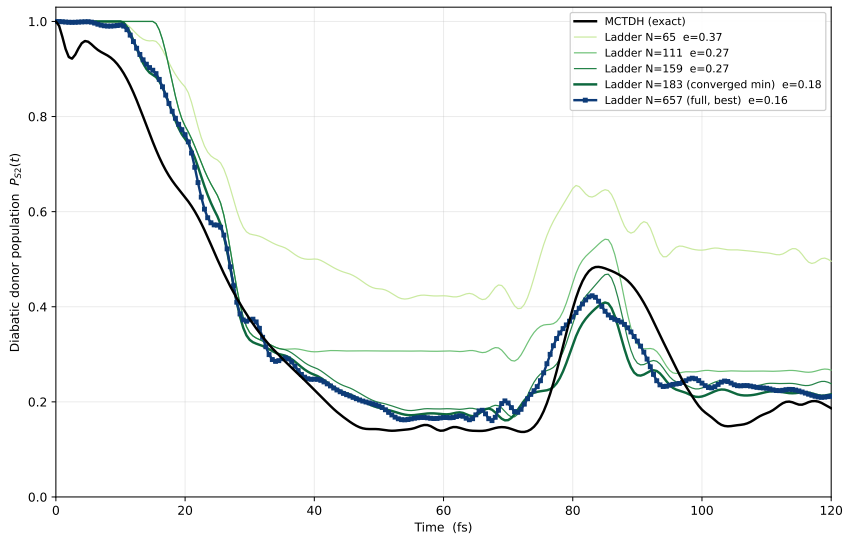
Results

AIMS results.



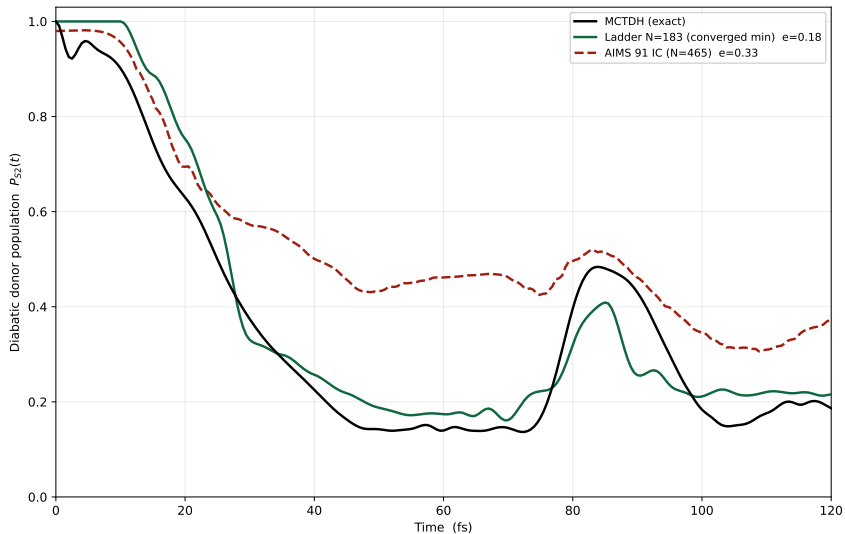
Results

AIMS with ladder operator results.



Results

Original AIMS and AIMS with ladder operator results.



Summary

- Extending AIMS with ladder operators and formulating it in the diabatic representation yields a compact basis that converges to the exact result (MCTDH) at the four-mode pyrazine model.
- Efficiency: Our method is $\sim 2\times$ more accurate at $\sim 40\%$ of the AIMS basis (183 vs 465) (0.178 vs 0.329).
- AIMS plateau: The AIMS error does not fall below ~ 0.33 regardless of the number of ICs or the spawn threshold — a structural limit of the single frozen Gaussian.

Future plans:

- Anharmonic potentials
- On the fly (Ab initio) dynamics
- Squeeze basis function $\Rightarrow$ back to the adiabatic picture

References

- [1] D. A. Fedorov, S. Seritan, B. S. Fales, T. J. Martínez, and B. G. Levine, J. Chem. Theory Comput. **16**, 5485–5498 (2020).
- [2] G. A. Worth, M. H. Beck, A. Jäckle, and H.-D. Meyer. The MCTDH Package, Version 8.2, (2000). H.-D. Meyer, Version 8.3 (2002), Version 8.4 (2007). O. Vendrell and H.-D. Meyer Version 8.5 (2013). Versions 8.5 and 8.6 contain the ML-MCTDH algorithm. See <http://mctdh.uni-hd.de>. Current versions: 8.4.24, 8.5.17 and 8.6.3 (2023).
- [3] The code was developed with the assistance of Claude (Anthropic).

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